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DATA STRUCTURES AND ALGORITHMS 2 (DSA711S) COURSE OUTLINE

Effective Start Date: 09 February 2026

STATEMENT ABOUT ACADEMIC HONESTY AND INTEGRITY

All staff and students of the Namibia University of Science and Technology (NUST), upon signing their employment contracts and registration forms, commit themselves to abide by the policies and rules of the institution. The core activity of NUST is learning and in this respect academic honesty and integrity is very important to ensure that learning is valid, reliable and credible.

NUST therefore does not condone any form of academic dishonesty, including plagiarism and cheating on tests and assessments, amongst other such practices. NUST requires students to always do their own assignments and to produce their own academic work, unless given a group assignment.

Academic Dishonesty includes, but is not limited to:

- Using the ideas, words, works or inventions of someone else as if it is your own work.
- Using the direct words of someone else without quotation marks, even if it is referenced.
  - Copying from writings (books, articles, webpages, other students' assignments, etc.), published or unpublished, without referencing.
  - Syndication of a piece of work, all or part of an assignment, by a group of students, unless the assignment was a legitimate group assignment.
- The borrowing and use of another person's assignment, with or without their knowledge or permission.
- Infringing copyright, including documents copied or cut and pasted from the internet.
  - Asking someone else to prepare an assignment for you or to write or sit an assessment for you, whether this is against payment or not.
  - Re-submitting work done already for another course or programme as new work, so-called self-plagiarism. Bringing notes into an examination or test venue, regardless of whether the notes were used to copy or not. Receiving any outside assistance in any form or shape during an examination or test.

All forms of academic dishonesty are viewed as misconduct under NUST Student Rules and Regulations. Students who make themselves guilty of academic dishonesty will be brought before a Disciplinary Committee and may be suspended from studying for a certain time or may be expelled. All students who are found guilty of academic dishonesty shall have an appropriate endorsement on their academic record, which will never be erased.

COURSE INFORMATION

COURSE CODE AND TITLE:

DSA711S - Data Structures and Algorithms 2

DEPARTMENT:

Computer Science

PROGRAMME:

07BCMS Bachelor of Computer Science

CONTACT HOURS:

Directed Hours:	**45**
Self-Directed Learning Hours:	**40**
Supervised Tutorials Hours:	**15**
Assessment Hours:	**20**
Total:	**120**

NQF LEVEL AND CREDIT:

NQF Level:	**7**
NQF Credits:	**12**

COURSE DESCRIPTION:

This course is designed to introduce students to advanced data structures and algorithms used to solve complex and computationally intensive problems.

PRE-REQUISITES:

Data Structures and Algorithms 1

COURSE EQUIVALENCIES:

None

COURSE DELIVERY METHODS:

Blended

The following communication tools will be used in this course:

Email, Discussion Board, Scheduled Chats, Unscheduled chats and online content

Course Format:

Ration of blended lecture and lab to self-Directed learning to assessment hours is 60:40:20**.**

EFFECTIVE DATE:

09th February 2026

LECTURER INFORMATION

Lecturer's name: Steven Tjiraso **(Coordinator),**

Coordinator's Contact Details:

Email: stjiraso@nust.na
Phone: +264 207 2097
Office location: HTTPS Building, Lower Campus Windheok West, Windhoek
Office hours: 07:30 – 16:30 and 17:15 – 21:20
Consultation hours: All lecturers are available for consultation with students (online consultation or via email). Lecturers should ensure that they have avail their consultation times and platform to students or by appointment.

STUDENT READINESS

Technology & Equipment Readiness:

NUST provides the technology needed for the course.

Student Commitments and Contact Times: See contact hours above

a) **Class Attendance:** Attend all classes, and be on time, and make appropriate notes. If a student fails to attend a class, it is their responsibility to catch-up on what was covered. The students have access to the recommended books from the library. If any special presentations are used in class, it will be made available and do not need to be copied by the student.

b) **Self-Directed Study:** After every class, and before the start of the next class, sort, complete, and annotate own lecture notes, and select further reading in case any particular item was not understood on the first attempt. A student must spend their notational hours revising, practicing or/and completing homework for the course. A student should take the reading list (prescribed and recommended reading seriously). Read all prescribed material to the extent that the student could write a one-page essay, summarising the content, without looking at the material again.

c) **Communications:** Maintain active communication with your group lecturer. Students are expected to use established channels of communication and platform to solve course-related issues on time. The end of the semester is not the best time to fix issues. Use the available time to deal with matters at hand.

Prescribed Reading:

Cormen, T. H. Leiserson, C. E, Rivest, R. L. & Stein, C (2009). Introduction to Algorithms (3rd ed.). 2009 Massachusetts Institute of Technology. ISBN 978-0-262-03384-8

Recommended Reading:

Knuth, D. E. (2015). The art of computer programming. Upper Saddle River: Addison-Wesley.
Sedgewick, R., & Wayne, K. D. (2017). Algorithms. Brantford, Ontario: W. Ross MacDonald Faculty Resource Services Library.
Skiena, S. S. (2012). The algorithm design manual. London: Springer.

Specific Learning Outcomes:

By the end of this course of study, students should be able to:

- Manipulate Advanced recursive data structures;
- Use various hashing techniques for faster data access;
- Compare algorithm efficiency using asymptotic analysis

Comprehensive Learning Outcome:

Analyse advanced recursive data structures as well as hashing techniques.

COURSE SCHEDULE:

This schedule may be adjusted if need be.

WEEK #	WEEK DATES	TOPICS	Activities: Discussion Forums, Reading Materials and Assessments
Week 1	**09 – 13 FEB 2026**	**Course Introduction and housekeeping**	• Introduction to the Programming Environment.
Week 2	**16 – 20 FEB 2026**	**Recursive functions and data structures** Introduction, Definitions and Basic terminologies of recursive structures	• Recursive programs •#understand short recursive programs
Week 3	**23 – 27 FEB 2026**	**Complexity analysis of recursive Structures** Recurrence relations	• Analysis • Analyse recursive programs empirically
Week 4	**02 – 07 MAR 2026**	**Balanced Search Trees** AVL trees	• Understand AVL tree operations using both iterative and recursive structures. • Analyse and optimize your programs using ideas from week 2-4
Week 5	**09 – 13 MAR 2026**	**Balanced Search Trees** AVL trees	• Understand AVL tree operations using both iterative and recursive structures. • Analyse and optimize your programs using ideas from week 2-4

Week 6	**16 – 20 MAR 2026**	**Balanced Search Trees** Red-Black trees	• Understand Red-Black tree operations using both iterative and recursive structures. • Analyse and optimize your programs using ideas from week 2-4
Week 7	**23 – 27 MAR 2026**	**Balanced Search Trees** B-trees 2-3 Search trees	• Understand B-tree and 2-3 tree operations using both iterative and recursive structures. • Analyse and optimize your programs using ideas from week 2-4
Week 8	**30 MAR – 2 APR 2026**	**SEMESTER BREAK**	
Week 9	**6 – 10 APR 2026**	**String Algorithms** Knuth-Morris-Pratt KMP) Matcher	• Analyze basic string algorithms in programs
Week 10	**13 – 17 APR 2026**	**Dynamic programming (DP)** Optimal substructure and memorization	• Optimize recursive solutions using DP concepts
Week 11	**20 – 24 APR 2026**	**Introduction to Hashing** Hash Tables, Hash Functions and their applications **Collision Handling techniques** Linear probing Chaining Double hashing	• Understand basic Hashing techniques
Week 12	**27 APR – 1 MAY 2026**	**Probabilistic Data Structures** Bloom Filter	
Week 13	**5 – 8 MAY 2026**	**Revision**	
Week 14	**11 – 15 MAY 2026**	**Consolidation of semester marks**	

IMPORTANT DATES:

NOTE: The following dates are subject to change based on the needs of the students at the lecturer's prerogative. Students will be notified ahead of time of any changes.

Sn	Assessment Type	Assessment Task	Date	Week
1		Distribution of Group Assignment Project	Friday, 23 March 2026	7
2	First Assessment	Test1	10 April 2026	9
3	Second Assessment	Test2	Friday, 24 April 2026	11
4	Third Assessment	Exercises (Individual Assignments)		From week 10 to week 12
		Last date to submit group assignment Project in MS Word documentation on elearning	17 April 2026	10
5		Supplementary	5 May 2026	13
6		Dates for Consolidation of Semester marks	11 – 15 May 2026	14
7	Examination		TBA	

ASSESSMENT AND EVALUATION:

The course will be assessed using a combination of Continuous Assessment and an end-of-semester examination. Continuous Assessment is made up of the following:

- Tests, Assignments/Projects, Exercises/Quizzes.

Students must obtain at least 40% to write the examination while a sub-minimum of 40% is required in the examination. The semester mark and examination mark shall be used jointly to determine the final mark in the ratio of 40% (semester mark) to 60% (examination mark). A minimum final mark of 50% is required to pass the course.

	Continuous Assessment (Semester Mark)	Weight
	Test 1	35%
	Test 2	35%
	Laboratory Exercises (Individual Assignments)	10%
	Group Assignment / Project	20%
A.	Total Semester Mark (minimum 40% required to write examination)	40%
B.	Examination Mark	60%
	Total Final Course Mark (A + B)	100%

Plagiarism and Deduction of Marks:

Marks will be deducted if students do not adhere to the rules of the University according to Rule AC3.2. All assignments should be submitted through Turnitin, the similarity software that is integrated into the MOODLE Learning Management System. If plagiarism is detected, marks should be deducted as follows:

It is the student's responsibility to be familiar with and adhere to the NUST's Policies. These Policies can be found in the NUST Prospectus or online at:

https://www.nust.na/sites/default/files/documents/eYearbook_P4_ComputingandInformatics.pdf

% Similarity Detected	% Marks Deduction
0 – 20	0
20 – 40	10
40 – 60	25
60 – 100	100

**** Please note that the % shown above is an average % for all Faculties and should be used as a Guideline.***

Assignments found with a similarity report above 20%, will not be allowed to apply for a remark or a recheck of marks. For students who fall into the 60 to 100% similarity group, Rule AC3.2 will apply, and the misconduct procedure will start.

Kindly note the difference between similarity and plagiarism.
The Turnitin software detects the similarity and assists staff and students with academic integrity. It should be used as follow:

- Pay attention to the repository setting. It should be set to "no repository" to avoid self plagiarism. Please note that the submitted work is compared with existing information in the database and other work that is available on the internet, such as electronic journals, websites, and internet resources.
- Turnitin should not be seen as a system that detects plagiarism, but it generates a similarity report. It identifies text matches and therefore guides staff and students to pay attention t o certain areas. A high text match does not indicate plagiarism, but rather a match with exis ting content in the database.
- A low level match, on the other hand might be plagiarised work which Turnitin could not detect because it is from a source that cannot be accessed or not available in the database.
- Set the error margin not to detect references.

Plagiarism is the act of taking someone else's work or ideas and passing them off as one's own and by failing to include quotations or give the appropriate citation by not adequately acknowledging an author of a source (NUST Policy on Plagiarism, 2020).

Lecturers will assess if the detected similarities are plagiarized materials
If it is found that the materials are plagiarised, then action as indicated in Rule AC3.2 will be taken.

Assessment Submission:

All assessments (assignments, projects, and activities) should be submitted on the eLearning platform to ensure that it goes through the Turnitin software to check for similarity.

Assessments should be typed and submitted in MS Word or PDF format. Please note that students should not submit an assignment as a JPG or any other image format, because Turnitin cannot read images.

Course Policies

General Academic Policies:

Failure to Pay Fees:

A student who fails to pay his/her fees may not be allowed to write the examination and if allowed, the results will be withheld until all outstanding fees are paid in full.

Important Student Services at NUST

There are a variety of services that you can use at NUST. These services are to your advantage – Use them!!! They include the following:

- Student Counseling and Career Development - Dean of Students Office.
- The Writing Centre and student academic problems – Centre for Teaching & Learning (CTL)
- Campus Health and Wellness Centre (CHWC) - Dean of Student's office/ NUST Clinic

AUTHORISATION:

This course is authorised for use by:

_____ _____

Head of Department **Date**

ACKNOWLEDGEMENT BY STUDENT

(To be completed by all students on the course, detached from the course outline and kept on record in the department)

I, _____, hereby acknowledge that I have received this course outline for (insert course title and code), and that I have familiarised myself with its content, in particular the statement about academic honesty and integrity. I agree to abide by the Policies and arrangements spelt out in this course outline.

_____ _____

Signature of Student **Date**